

Derivatives — group rounds

Lessons 3–4, 7–12 · Quarter I · 40 min · 19 questions · 84 points · groups of 3–4

Four rounds, each timed and worth more than the last. The tag on every question names the topic it comes from. Work on paper, hand in one sheet for the group. An answer with no method scores half.

LESSON CLOCK · 40 MINUTES

4' Form groups, read the rules	5' Round A	8' Round B	9' Round C	8' Round D	6' Answers on the board, scores
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1. Groups of 3–4. One answer sheet per group, with every member’s name on it.
2. Rounds are timed and called by the teacher. When a round is called, pens down and move on — an unfinished round scores what it has.
3. In rounds C and D any member may be asked to explain the method. If nobody can, the group scores half for that question, however correct the answer is.
4. No calculators. Leave surds and π exact, and factorise every derivative.

Round A · Easy

5'2 pts each

Three topics, eight questions — speed round

1

T2

Evaluate $\lim$ as $x \rightarrow -2$ of $\frac{x^2 + 5x + 6}{x + 2}$.

1

The numerator factorises as $(x + 2)(x + 3)$, so after cancelling the limit is $x + 3 \rightarrow 1$.

2

T3

Differentiate $y = 5\sqrt{x} - \frac{3}{x^2}$.

$\frac{5}{2\sqrt{x}} + \frac{6}{x^3}$

Rewrite as $5x^{1/2} - 3x^{-2}$, then $\frac{5}{2}x^{-1/2} + 6x^{-3}$.

3

T2

Evaluate $\lim$ as $x \rightarrow 3$ of $\frac{x^2 - 9}{x^2 - 2x - 3}$.

$\frac{3}{2}$

Both factorise through $(x - 3)$: $\frac{(x - 3)(x + 3)}{(x - 3)(x + 1)} = \frac{x + 3}{x + 1} \rightarrow \frac{6}{4}$.

4

T2

Evaluate $\lim$ as $x \rightarrow 0$ of $\frac{\sqrt{x + 1} - 1}{x}$.

$\frac{1}{2}$

Conjugate: the numerator becomes x , leaving $\frac{1}{\sqrt{x + 1} + 1}$.

5

T3

Differentiate $y = \frac{2x^3 - 5}{x^2}$.

$2 + \frac{10}{x^3}$

Divide first: $y = 2x - 5x^{-2}$, so $y' = 2 + 10x^{-3}$.

6

T3

Differentiate $y = x\sqrt{x}$.

$\frac{3}{2}\sqrt{x}$

$x\sqrt{x} = x^{3/2}$, so $y' = \frac{3}{2}x^{1/2}$.

7

T4

Differentiate $y = \frac{1}{(2x - 7)^3}$.

$-\frac{6}{(2x - 7)^4}$

$y = (2x - 7)^{-3}$, so $y' = -3(2x - 7)^{-4} \times 2$.

8

T4

Differentiate $y = \sqrt[3]{3x + 1}$.

$\frac{1}{(3x + 1)^{2/3}}$

$y = (3x + 1)^{1/3}$, so $y' = \frac{1}{3}(3x + 1)^{-2/3} \times 3$ — the 3s cancel.

Round B · Medium

8'4 pts each

Rewrite before you differentiate

9

T4

Differentiate $y = \sqrt{1 - 4x^3}$.

$-\frac{6x^2}{\sqrt{1 - 4x^3}}$

Chain rule: $\frac{1}{2\sqrt{1 - 4x^3}} \times (-12x^2)$; the 2 cancels.

10

T2

Evaluate $\lim$ as $x \rightarrow 2$ of $\frac{x^3 - 8}{x - 2}$.

12

$x^3 - 8 = (x - 2)(x^2 + 2x + 4)$, so the limit is $4 + 4 + 4$.

11

T2

Evaluate $\lim$ as $x \rightarrow 1$ of $\frac{1}{x - 1} - \frac{2}{x^2 - 1}$.

$\frac{1}{2}$

Neither part has a limit on its own. Over a common denominator:
 $\frac{(x + 1) - 2}{(x - 1)(x + 1)} = \frac{x - 1}{(x - 1)(x + 1)} = \frac{1}{x + 1}$.

12

T3

Differentiate $y = (x + 1)(x - 2)(x + 3)$.

$3x^2 + 4x - 5$

Expanding is quicker than two product rules: $y = x^3 + 2x^2 - 5x - 6$.

13

T3

Differentiate $y = \frac{x^2 + 1}{x^2 - 1}$.

$-\frac{4x}{(x^2 - 1)^2}$

Quotient rule; the numerator is $2x(x^2 - 1) - (x^2 + 1)2x = 2x(-2) = -4x$.

14

T4

Differentiate $y = (1 - 2x)^6(x + 3)$ and factorise.

$-7(1 - 2x)^5(2x + 5)$

Product rule with the chain rule on the first factor:
 $-12(1 - 2x)^5(x + 3) + (1 - 2x)^6$. Take out $(1 - 2x)^5$; what is left is
 $-12(x + 3) + (1 - 2x) = -14x - 35$.

15 T3 Differentiate $y = \frac{x^2 + 1}{x + 2}$ and find every x at which the tangent is horizontal.

$y' = \frac{x^2 + 4x - 1}{(x + 2)^2}$; $x = -2 \pm \sqrt{5}$

Quotient rule: the numerator is $2x(x + 2) - (x^2 + 1) = x^2 + 4x - 1$. A fraction is zero only when its numerator is, so solve $x^2 + 4x - 1 = 0$; completing the square gives $(x + 2)^2 = 5$. Two places, and neither is a whole number — groups that expect a tidy root stop too early.

16 T2 Evaluate $\lim_{x \rightarrow 4} \text{ of } \frac{\sqrt{x} - 2}{x^2 - 16}$.

$\frac{1}{32}$

Factorise the denominator as $(x - 4)(x + 4)$ and multiply above and below by $\sqrt{x} + 2$. The numerator becomes $x - 4$, which cancels, leaving

$\frac{1}{(\sqrt{x} + 2)(x + 4)} \rightarrow \frac{1}{4 \times 8}$.

17 T3 Differentiate $y = \frac{x + 1}{\sqrt{x}}$ for $x > 0$, and give the answer as a single fraction.

$\frac{x - 1}{2x\sqrt{x}}$

The quotient rule works, but rewriting is far quicker: $y = x^{1/2} + x^{-1/2}$, so $y' = \frac{1}{2}x^{-1/2} - \frac{1}{2}x^{-3/2}$. Taking out $\frac{1}{2}x^{-3/2}$ leaves $x - 1$, giving $\frac{x - 1}{2x\sqrt{x}}$.

Check: at $x = 4$ this is $\frac{3}{16}$, and the curve is rising there.

18 T4 Differentiate $y = \sqrt{\frac{2x + 1}{2x - 1}}$ for $x > \frac{1}{2}$.

$-\frac{2}{(2x - 1)\sqrt{4x^2 - 1}}$

Chain rule on $u^{1/2}$ with $u = \frac{2x + 1}{2x - 1}$. The quotient rule gives

$\frac{du}{dx} = \frac{2(2x - 1) - 2(2x + 1)}{(2x - 1)^2} = -\frac{4}{(2x - 1)^2}$, so $y' = \frac{1}{2\sqrt{u}} \times (-\frac{4}{(2x - 1)^2})$.

Writing $\sqrt{u} = \frac{\sqrt{2x + 1}}{\sqrt{2x - 1}}$ and tidying gives the answer. The derivative is negative everywhere on the domain, which matches a decreasing curve.

Round D · Very hard 8' 16 pts each

Every part needed for full marks

19 T3 T4 (a) Show that $y = \frac{x}{\sqrt{x^2 + 4}}$ has $y' = \frac{4}{(x^2 + 4)^{3/2}}$. (b) Hence find the equation of the tangent at $x = 0$. (c) Explain why the curve has no stationary point.

(b) $y = \frac{x}{2}$ (c) y' is never zero

(a) Quotient rule: $[\sqrt{x^2 + 4}] - x \cdot \frac{x}{\sqrt{(x^2 + 4)}}]/[x^2 + 4]$. Multiply above and below by $\sqrt{x^2 + 4}$: the numerator becomes $(x^2 + 4) - x^2 = 4$ and the denominator $(x^2 + 4)^{3/2}$.

(b) $y(0) = 0$ and $y'(0) = \frac{4}{8} = \frac{1}{2}$, so the tangent is $y = \frac{1}{2}x$.

(c) The numerator of y' is the constant 4, so $y' \neq 0$ for every x — the curve is increasing everywhere and never levels off.

TOPIC COVERAGE

TOPIC	QUESTIONS	WHICH
T2 The limit of a function · lessons 3–4	6 1, 3, 4, 10, 11, 16	
T3 The rules of differentiation · lessons 7–9	8 2, 5, 6, 12, 13, 15, 17, 19	
T4 The derivative of a composite function · lessons 10–12	6 7, 8, 9, 14, 18, 19	

SCORE

ROUND	QUESTIONS	EACH	ROUND TOTAL
Round A · Easy	8	2	16
Round B · Medium	6	4	24
Round C · Hard	4	7	28
Round D · Very hard	1	16	16
Total	19		84